

Yaowen Lu, PhD, CFA

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WORK EXPERIENCE

Quantitative Developer

Jacobi Strategies Pty Ltd

Dec. 2022 – Present

Brisbane, Australia

- Developed economic scenario generation models using stochastic and regime-switching frameworks to simulate multiple paths for asset returns, yields, and macroeconomic variables
- Supported the investment engineering team in designing and maintaining a multi-asset portfolio platform, focusing on scenario simulation and portfolio optimization
- Applied mathematical and statistical methods in collaboration with investment specialists to quantify real-world portfolio design problems and implement practical solutions
- Collaborated with the investment engineering team to design and build mathematical models as computer applications with suitable user interfaces and visualization tools

Adjunct Lecturer

The University of Queensland, School of Mathematics and Physics

Sep. 2026 – Present

Brisbane, Australia

- Committed to enhancing academia–industry collaboration through knowledge exchange, professional engagement, and the integration of industry insights into teaching and research

Casual Tutor

The University of Queensland, School of Mathematics and Physics

Feb. 2020 – Nov. 2022

Brisbane, Australia

- Course Title: Computation in Financial Mathematics, focusing on numerical methods for solving financial differential equations
- Course Title: Financial Calculus, covering derivative pricing and stochastic calculus

Quantitative Analyst – Capital Management

China CITIC Bank, Department of Assets and Liabilities

Apr. 2013 – Jul. 2017

Beijing, China

- Won the Employee of the Year award in 2015
- Investigated capital rules and regulations under the framework of BASEL III and IFRS 9
- Calculated and reported capital adequacy ratio data to executive management every month
- Conducted measurement of risk-weighted assets (RWA), including credit, market, and operational RWA
- Performed capital structure analysis, capital adequacy projection, and capital measurement training
- Engaged in capital information disclosure, regulatory submission, and external audits
- Coordinated establishment of capital adequacy rate measurement and monitoring system

Management Trainee – Strategies Analysis

China Merchant Bank, Department of Planning and Finance

Apr. 2011 – Apr. 2013

Shanghai, China

- Rotated through departments of sales, marketing, risk management, operation, and financial planning
- Performed data analysis to optimize product innovation and marketing strategy, including but not limited to client profile analysis, behavior analysis, profit, and loss analysis

Analytic Consultant – Full-time Intern

Fair Isaac (FICO) Information Technology Co. Ltd

Oct. 2010 – Mar. 2011

Beijing, China

- Wrangled massive retail banking data for Industrial and Commercial Bank of China
- Built scorecard models for retail banking services and gained hands-on experience with big data

EDUCATION

Master of Science | *Financial Engineering*

WorldQuant University

Jun. 2025 – Present

Online

- Relevant courses: Financial Data, Financial Markets, Financial Econometrics, Machine Learning in Finance

Doctor of Philosophy | *Computational Finance*

The University of Queensland, School of Mathematics and Physics

Aug. 2017 – Nov. 2022

Brisbane, Australia

- Won First Place of 1st-time presenter for mathematics at SMP Poster Day 2019
- Won the Australian government research training program scholarship

- Dissertation: Numerical methods for Guaranteed Minimum Withdrawal Benefits

Master of Science | *Mathematics*

Sep. 2008 – Jan. 2011

Beihang University, School of Mathematics and Systems Science

Beijing, China

- Supported by First Class Scholarship (7 scholarships in 42 applicants), GPA: 3.4/4.0
- Won Second Prize in 2010 National Postgraduate Mathematical Contest in Modeling
- Relevant courses: Numerical Methods for Partial Differential Equations, Numerical Approximation

Bachelor of Science | *Information and Scientific Computing*

Sep. 2004 – Jul. 2008

Beihang University, School of Mathematics and Systems Science

Beijing, China

- Graduated with Excellent Graduate Award (5 scholarships in 120 applicants), GPA: 3.5/4.0
- Received merit-based National Endeavor Scholarship (4 scholarships in 120 applicants)
- Relevant Courses: Calculus, Advanced Algebra, Probability and Statistics, Numerical Analysis

RESEARCH EXPERIENCE

Numerical Methods for the Guaranteed Minimum Withdrawal Benefits

Aug. 2017 – Nov. 2022

The University of Queensland, Doctoral Research Project

Brisbane, Australia

- Solved the pricing problem of Guaranteed Minimum Withdrawal Benefits (GMWB) under jump diffusions
- Extended the pricing framework to stochastic interest rates using a semi-Lagrangian discretization
- Incorporated more stochastic factors into the framework, e.g. stochastic volatility and multiple subaccounts
- Applied deep neural network methods for multi-factors guaranteed benefits pricing problems

Multi-domain Numerical Methods in Computational Fluid Dynamics

Sep. 2008 – Mar. 2011

Beihang University, Postgraduate Research Project

Beijing, China

- Compared the weighted essentially non-oscillatory (WENO) methods with Runge-Kutta discontinuous Galerkin (RKDG) methods for hyperbolic conservation laws
- Combined them together to solve practical problems with irregular boundary conditions

PUBLICATIONS

- An ϵ -monotone Fourier method for GMWB as a continuous impulse control problem, Yaowen Lu, Duy-Minh Dang, Peter Forsyth, George Labahn, submitted [Preprint]
- A semi-Lagrangian ϵ -monotone Fourier method for continuous withdrawal GMWBs under jump-diffusion with stochastic interest rate, Yaowen Lu, Duy-Minh Dang, Numerical Methods for Partial Differential Equations. 10.1002/num.23075. [Journal Version]
- A pointwise convergent numerical integration method for Guaranteed Lifelong Withdrawal Benefits under stochastic volatility, submitted [Preprint]
- Multi-Domain Hybrid RKDG and WENO methods for Hyperbolic Conservation Laws, Jian Cheng, Yaowen Lu, Tiegang Liu, SIAM Journal on Scientific Computing, Vol. 35, No. 2, pp. A1049-A1072 [Journal Version]

PRESENTATIONS

- The 67th Annual Meeting of the Australian Mathematical Society, 2023, Brisbane, Australia
- The Canadian Applied and Industrial Mathematics Society (CAIMS) Annual Meeting 2021, Online
- The Quantitative Methods in Finance 2019 Conference, Sydney, Australia
- The 20th INFORMS Applied Probability Society Conference, 2019, Brisbane, Australia

MISCELLANEOUS

Certificates: CFA Charterholder, DeepLearning.AI TensorFlow Developer

Languages: English (Professional proficiency), Mandarin (Native)

Technical Skills:

- Python: Proficient with NumPy, SciPy, CVXPY, pandas, scikit-learn, statsmodels, Plotly, backtrader, etc.
- C++: Developed Fourier-based numerical methods for optimal control problems
- MATLAB: Developed multiple numerical methods to financial applications
- SQL/NoSQL: Hands-on experience in data analysis and data modeling
- Databricks: Experience working with Databricks for large-scale data processing

- Linux: Proficient with Linux command-line workflows
- DevOps: Working knowledge of Git and Docker
- Apps: Practical experience with Dash and React for interactive applications

Document Creation: Microsoft Office Suite, L^AT_EX, Markdown, HTML

Leisure Activities: Fishing, kayaking, dog walking, table tennis, basketball, and e-gaming